

Malcolm Cephas

Backend Developer | AI Systems Engineer | ECE Graduate

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SUMMARY

Backend Developer and AI Systems Intern with experience building Java Spring Boot applications, AI-powered enterprise solutions, and secure distributed systems. Skilled in Java, Python, Spring Boot, REST APIs, SQL, MCP-based AI integrations, and LLM workflows. Experienced in developing backend services, conversational AI applications, and cryptography-based access control systems.

EDUCATION

Bachelor of Engineering – Electronics and Communication Engineering

2022 - 2026

Dayananda Sagar College of Engineering, Bangalore · 8.2 CGPA

EXPERIENCE & INTERNSHIPS

AI Intern

Jan 2026 - Present

Sensei Technologies · Onsite, Bangalore

- Developed MCP-based backend services using Java, Spring Boot, Spring AI, and MCP server/client architecture.
- Integrated MCP services with Claude Desktop, Groq, OpenRouter, and Ollama to enable conversational system interaction and tool orchestration.
- Developed an AI chatbot supporting persistent chat memory, multi-chat functionality, and conversation summarization.
- Built metadata-driven AI workflows and schema-aware SQL generation using structured tool definitions, prompt engineering, and database metadata.
- Researched and optimized LLM workflows through token reduction, jq-based filtering, and local model execution using Ollama.
- Evaluated locally hosted LLMs including SmolLM3, Qwen3, and Llama 3 for application integration.

Java Development Intern

Oct 2025 – Dec 2025

Elevate Labs · Remote, Bangalore

- Developed six Java applications covering CRUD operations, file handling, object-oriented programming, data structures, and GUI development.
- Built command-line applications including Student Management, Library Management, Notes Manager, and Bank Account Management systems.
- Developed a Java Swing GUI-based To-Do List with task creation, deletion, persistence, and automatic loading of saved tasks.
- Applied object-oriented programming and core Java concepts to design modular application logic and solve practical programming tasks.

PROJECTS

IoT Systems

- Designed an ECC-based access control system for distributed IoT networks, securing data access across resource-constrained devices.
- Applied I2C and UART communication protocols within the IoT architecture for data exchange between embedded nodes.
- Implemented authentication and access-control logic using C++, Java, and Python.
- Applied Attribute-Based Encryption (ABE) techniques to enable controlled and selective access to protected resources.
- Awarded “Best Project of Social Relevance” at DSCE Open Day 2026.

E-Commerce System with MCP Server *MCP · Spring AI · Java · Spring Boot · MySQL · Hibernate · React*
Integration

- Developed a full-stack e-commerce application with a ReactJS frontend and Java Spring Boot backend.
- Built REST APIs in Spring Boot to handle application logic and communication between the frontend and backend.
- Implemented MySQL database persistence using Hibernate.
- Integrated an MCP server with custom tools and prompts to enable conversational interaction with the e-commerce system.
- Connected the application with Spring AI and LLM services to support AI-powered functionality.

Expense Splitter *Spring Boot · JavaFX · Java · Greedy Algorithms · REST APIs*

- Developed a full-stack expense management application using a Java Spring Boot server and JavaFX desktop client, with a greedy algorithm to minimize settlement transactions.
- Integrated a live multi-currency exchange-rate API with offline JSON-based caching.

SKILLS

Java	Python	C++	JavaScript	SQL	Spring Boot	Spring AI	MCP	REST APIs	React.js	MySQL
PostgreSQL	Hibernate	AWS	Docker	Git	GitHub	Maven	Gradle	Postman	Machine Learning	
NumPy	Pandas	PyTorch	Cryptography	Network Security	ECC					

CERTIFICATIONS

- Model Context Protocol (MCP) — KodeKloud
- AI Infrastructure & Operations — NVIDIA / Coursera
- AWS Solutions Architect, Developer & SysOps — Udemy
- Introduction to Software Engineering — IBM / Coursera
- An Introduction to Cryptography — University of Leeds / Coursera

LANGUAGES

English, Kannada, Tamil

